Specialist Hospital Service (Hospital Patient Administration System)

Administrative Management Patient Pathway Coordinators Outpatient Bookings

Team Medical Secretaries Consultant / Clinical

Team Finance

Team

Treatment & Chemotherapy Bookings

Team
Authorised
treatment booking
request received

Validate the authorised
request and confirm
resource requirements

Request treatment
appointments from
external clinical services
External
treatment

capacity
confirmed?

Keep the booking
pending, notify the
responsible team and
record the retry

Determine the funding
route: hospital funded,
approved insurer or
patient payable
Funding approved
or no payment
required?

Record the funding
organisation,
authorisation reference,
approved amount and
limitations

Request funding
approval or additional
payment

Advance payment
required before
confirmation?

Send the secure
payment request to the
External Payment
Service Provider
Payment outcome
returned by the
payment service?

Record the payment
reference, amount and
date against the patient
account

Notify the patient and
the responsible team and
allow a further attempt
without a duplicate
booking

Mark the transaction for
investigation and do not
raise a second payment

Authorise urgent
treatment without
confirmed payment and
record the reason

Resolve the outstanding
transaction with the
payment provider or
funding organisation

Refer the unconfirmed
payment to the Finance
Team for later resolution

Confirm the treatment
schedule and issue the
patient instructions

Deliver the treatment
cycle and record clinical
administration
Further
treatment cycles
planned?

Request the pre-cycle
blood test and clinical
assessment

Review the results and
confirm whether the
patient is fit to continue
Patient fit
to continue
as

planned?

Raise and authorise a
Treatment Modification
Request

Record the urgent
postponement for patient
safety and obtain
retrospective
authorisation

Modification
affects an
existing charge,
funding
approval or
payment?

Review the financial
implications and update
the funding or payment
record

Apply the authorised
change and update the
treatment schedule

Prepare the clinic letter

Approve the clinical
content of the clinic letter

Clinical
content
approved?

Perform the
administrative checks
and confirm the intended
recipients
Suspected clinical
error in the letter?

Does the patient
require an
accessible format,
translation or a
representative?

Produce the
correspondence in the
required accessible
format or language

Distribute the clinic
letter through the
approved communication
channels

Issue a weekly reminder
to the responsible
consultant
Wait one week
before the
next reminder

Letter
outstanding
for more than
one month?

Escalate to the
Administrative Manager,
who contacts the
consultant
Letter
outstanding for
more than three
months?

Escalate to the higher
management team
Delayed
correspondence
escalated

Request a follow-up
appointment within the
clinically specified period

Arrange the follow-up
appointment using the
External Scheduling
Service
Appointment
found within the
requested period?

Highlight the case and
refer it to the relevant
pathway team
Follow-up outside
the clinical
timescale
escalated

Confirm the follow-up
appointment and notify
the patient

Record the cancellation
or non-attendance and
determine the next step
Does the
cancelled
appointment
carry a payment?

Determine whether the
payment should be
retained, transferred or
refunded

Send the approved
refund request to the
External Payment
Service Provider

Record the refund result
against the patient
account

Offer a further
appointment or refer the
pathway for clinical
review Appointment not
attended: pathway
reviewed

Complete the pathway
and discharge the patient
Patient
pathway
completed

ABSTRACTION LEVEL 2 - CROSS-FUNCTIONAL STRATEGIC COLLABORATION. Scope: treatment funding and payment, treatment delivery and review cycles, authorised treatment modification, clinical correspondence and pathway monitoring, follow-up and aftercare,
patient enquiry handling, and management reporting. Three processes are shown because they are genuinely independent: the patient pathway, enquiries (a patient may call at any time) and reporting (a monthly cycle, not a pathway step).

LINK TO THE DETAILED OPERATIONAL MODEL. The funding and payment section decomposes into the operational model O1_Operational_Merged: T_DetermineFunding > O1 T_DetermineFundingRoute; T_RecordFundingApproval > O1 T_RecordFundingApproval;
T_RequestSecurePayment > O1 T_SendSecurePaymentRequest (Zeebe job request-secure-payment, message flows MF_6 and MF_7); T_HandlePaymentFailure > O1 T_HandlePaymentFailure; T_MarkForInvestigation > O1 T_MarkForInvestigation; T_AuthoriseUrgentTreatment
> O1 T_AuthoriseUrgentTreatment. Treatment delivery, cycle review, modification, correspondence, enquiries, follow-up, aftercare and reporting are modelled here but are NOT implemented in the initial release: those activities are the gap between the
wider business process and the built solution.

EXCEPTION DESIGN REQUIRED BY THE CASE STUDY. An unavailable external service leaves the booking pending with a recorded retry and no duplicate appointment (G_CapacityConfirmed, T_KeepPending). A payment that is declined, cancelled or incomplete
allows a further attempt without creating a duplicate booking (T_HandlePaymentFailure). A payment taken by the provider whose confirmation was not returned is marked for investigation and is NEVER automatically re-requested (T_MarkForInvestigation).
Urgent treatment may proceed without confirmed payment where delay would create a patient risk, provided the reason is recorded and referred to the Finance Team (T_AuthoriseUrgentTreatment, T_ReferUnconfirmedPayment). Clinical staff may not approve
refunds without the required financial authority, and the Finance Team may not decide whether treatment is clinically necessary (T_RefundDecision).

PATHWAY MONITORING AND ESCALATION. A non-interrupting boundary timer starts pathway monitoring if a clinic letter is not approved within seven days, while the Consultant still owes the letter. The Patient Pathway Coordinators issue weekly reminders;
after one month the case is escalated to the Administrative Manager, who contacts the consultant; after three months it is escalated to the higher management team. Because the timer is non-interrupting and is cancelled when the letter is approved,
repeated reminders are not generated unnecessarily after a letter has been completed, and a full history of reminders, responses and escalations is retained.

Patient cancels,
declines or does
not attend

Clinic letter
not approved
within seven
days

Not confirmed

Confirmed

Retry without
duplicate booking

Approval required

Approved or not
required

Approval recorded

No advance
payment

Advance payment
required

Payment
confirmed

Declined,
cancelled
or

incomplete

Further attempt
authorised

Confirmation not
returned

Urgent
clinical need

Treatment
proceeds

Refer for financial
resolution

Further No further cycles
cycles
planned

Fit to continue

Planned change
required

Immediate
postponement
required

Authorised
retrospectively

No
financial
impact

Financial impact

Funding
record updated

Updated schedule

Not approved

Approved

Returned to the
Consultant

No clinical error Standard format

Accessible format,
translation or
representative
required

Seven days
elapsed

Less than
one month

Less than
three months

More than
one month

More than
three months

Within period

Not within period

Payment involved

Refund approved

No payment
involved

Payment retained
or transferred

Hospital Patient Administration System - patient enquiry handling

Finance

Team Clinical Nurse Specialist

Team Call Handling

Team
Patient or
representative
enquiry received

Record, classify and
prioritise the enquiry

What type of
enquiry is it?

Answer or transfer the
administrative enquiry to
the responsible team

Answer the payment or
funding enquiry using
authorised information

Provide clinical advice
through an authorised
clinical professional

Highlight and escalate
the urgent clinical
concern immediately

Record the response,
responsible team and
resolution status
Enquiry resolved
and closed

Administrative

Financial

Clinical Urgent
clinical
concern

Hospital Patient

Administration System

- management reporting

Hospital Management

Monthly
reporting period
reached

Generate referral,
waiting time,
correspondence,
funding, payment and
pathway reports

Review performance
reports and agree
corrective action
Reports published
and corrective
action agreed

Patient / Authorised Representative External Clinical Services (treatment, laboratory, imaging)

Funding Organisation / Insurer

External Payment Service Provider

External Correspondence Service

External Scheduling Service

TTreatment capacity response reatment capacity request

Blood test and assessment
request

Funding approval request

Funding approval response

Secure payment request

Payment status

Refund request

Clinical correspondence

Follow-up appointment request

Follow-up appointment response

Patient cancellation or
non-attendance

Patient enquiry